A FINITE FIELD EXTENSION IS FINITELY GENERATED, AND NOT CONVERSELY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The one-line reason finite generation is strictly weaker than finiteness for field extensions.

Editorial note. The label lemma-finite-finitely-generated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- A basis of F over E spans F as an E-vector space, and F = E(alpha_1..alpha_n) for any spanning set
- The immediately preceding worked example example-degree-rational-function-field (source lines 682-704): the powers t^n, n in Z, are linearly independent over k, so k(t)/k is not finite

Source reference. Original reference 09G6: sources/stacks-fields/pinned-original-source.tex, source lines 683–683. Prerequisite pointer invoked at 717; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *A finite extension of fields is a finitely generated field extension. The converse is not true.*

Proof. Let F/E be a finite extension of fields. Let $\alpha_1, \ldots, \alpha_n$ be a basis of F as a vector space over E. Then $F = E(\alpha_1, \ldots, \alpha_n)$ hence F/E is a finitely generated field extension. The converse is not true as follows from Example 09G6. □

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